Week #10: Numerical Integration

Numerical Methods: Mathematical Foundations

June 22, 2025

Overview

Numerical integration, or quadrature, addresses the fundamental problem of approximating definite integrals when analytical evaluation is impossible or impractical. This week we develop the mathematical theory underlying numerical integration methods, from basic Newton-Cotes formulas to sophisticated adaptive and Gaussian quadrature techniques.

The central challenge in numerical integration is to balance accuracy and computational efficiency. We explore how different quadrature rules achieve this balance through careful choice of evaluation points and weights, and examine the mathematical principles that determine convergence rates and error bounds.

Our analysis emphasizes the mathematical foundations of quadrature theory, the role of polynomial approximation in integration methods, and the error analysis that guides method selection and adaptive strategies.

1 Fundamental Concepts

1.1 The Quadrature Problem

Definition. Numerical Quadrature

The numerical quadrature problem is to approximate the definite integral:

$$I[f] = \int_a^b f(x)dx$$

using a quadrature rule of the form:

$$Q[f] = \sum_{i=0}^{n} w_i f(x_i)$$

where x_i are the quadrature nodes and w_i are the quadrature weights.

Definition. Degree of Precision

A quadrature rule has degree of precision d if it integrates all polynomials of degree $\leq d$ exactly, but fails for some polynomial of degree $d+1$.

1.2 Error Analysis Framework

Theorem 1.1 (General Error Formula)**.** *If the quadrature rule $Q[f]$ has degree of precision d and $f \in C^{d+1}[a,b]$, then:*

$$I[f] - Q[f] = \int_a^b f(x)K(x)dx$$

where $K(x)$ is the Peano kernel associated with the quadrature rule.

2 Newton-Cotes Formulas

Newton-Cotes formulas use equally spaced nodes and are derived by integrating interpolating polynomials.

2.1 Closed Newton-Cotes Formulas

Definition. Closed Newton-Cotes Formulas

Closed formulas include both endpoints a and b as nodes. For $n+1$ equally spaced points $x_i = a + ih$ where $h = (b-a)/n$:

$$\int_a^b f(x)dx \approx \sum_{i=0}^{n} w_i f(x_i)$$

2.1.1 Trapezoidal Rule

Definition. Trapezoidal Rule

With $n = 1$:

$$\int_a^b f(x)dx \approx \frac{b-a}{2}[f(a) + f(b)]$$

Theorem 2.1 (Trapezoidal Rule Error)**.** *If $f \in C^2[a,b]$, then:*

$$\int_a^b f(x)dx - \frac{b-a}{2}[f(a) + f(b)] = -\frac{(b-a)^3}{12} f''(\xi)$$

for some $\xi \in (a,b)$.

2.1.2 Simpson's Rule

Definition. Simpson's Rule

With $n = 2$:

$$\int_a^b f(x)dx \approx \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$

Theorem 2.2 (Simpson's Rule Error)**.** *If $f \in C^4[a,b]$, then:*

$$\int_a^b f(x)dx - \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right] = -\frac{(b-a)^5}{2880} f^{(4)}(\xi)$$

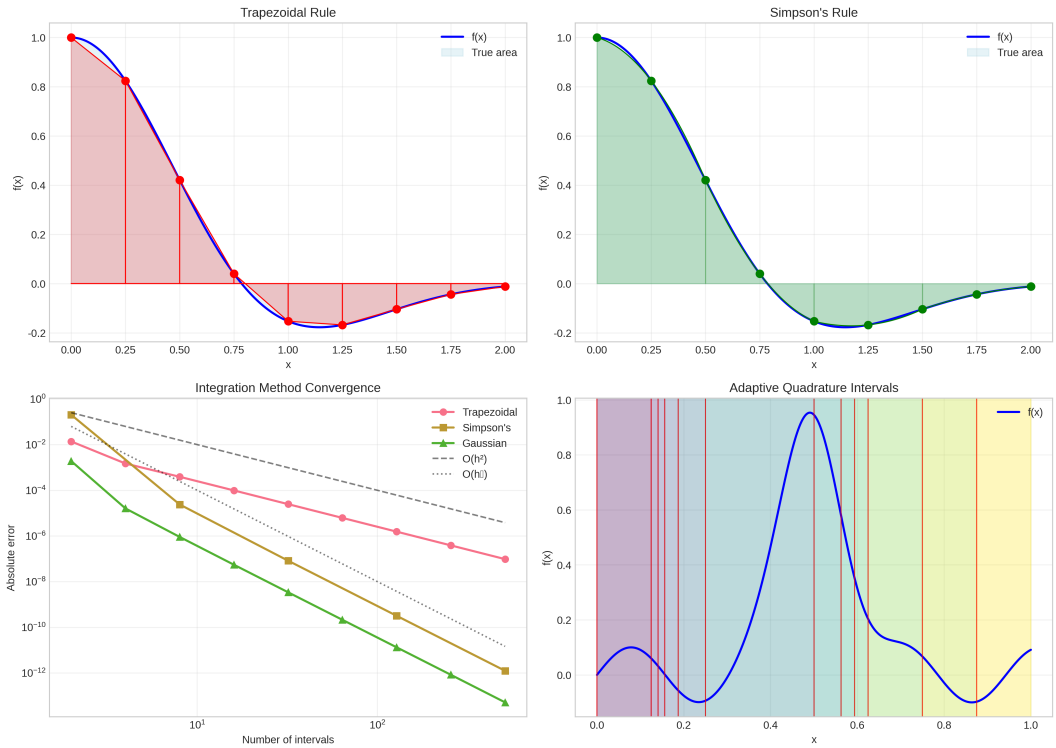

Figure 1: Numerical integration methods comparison. The left panel shows the geometric interpretation of different quadrature rules. The right panel illustrates the convergence behavior of composite rules and adaptive strategies.

2.2 Open Newton-Cotes Formulas

Open formulas exclude the endpoints, useful when the integrand has singularities at the boundaries.

Definition. Midpoint Rule

The simplest open formula:

$$\int_a^b f(x)dx \approx (b-a)f\left(\frac{a+b}{2}\right)$$

Theorem 2.3 (Midpoint Rule Error)**.** *If $f \in C^2[a,b]$, then:*

$$\int_a^b f(x)dx - (b-a)f\left(\frac{a+b}{2}\right) = \frac{(b-a)^3}{24}f''(\xi)$$

3 Composite Quadrature Rules

For improved accuracy, divide the integration interval into subintervals and apply basic rules to each.

3.1 Composite Trapezoidal Rule

Definition. Composite Trapezoidal Rule

Divide $[a,b]$ into n equal subintervals of length $h = (b-a)/n$:

$$\int_a^b f(x)dx \approx \frac{h}{2}\left[f(a) + 2\sum_{i=1}^{n-1} f(a+ih) + f(b)\right]$$

Theorem 3.1 (Composite Trapezoidal Error)**.** *If $f \in C^2[a,b]$, then:*

$$\left| \int_a^b f(x)dx - T_n[f] \right| \leq \frac{(b-a)h^2}{12} \max_{x\in[a,b]} |f''(x)|$$

where $T_n[f]$ is the composite trapezoidal approximation.

3.2 Composite Simpson's Rule

Definition. Composite Simpson's Rule

Divide $[a,b]$ into $2n$ equal subintervals:

$$\int_a^b f(x)dx \approx \frac{h}{3}\left[f(a) + 4\sum_{i=1}^{n} f(a+(2i-1)h) + 2\sum_{i=1}^{n-1} f(a+2ih) + f(b) \right]$$

Theorem 3.2 (Composite Simpson's Error)**.** *If $f \in C^4[a,b]$, then:*

$$\left| \int_a^b f(x)dx - S_n[f] \right| \leq \frac{(b-a)h^4}{180} \max_{x\in[a,b]} |f^{(4)}(x)|$$

4 Gaussian Quadrature

Gaussian quadrature achieves maximum degree of precision by optimally choosing both nodes and weights.

4.1 Optimal Quadrature Theory

Theorem 4.1 (Gaussian Quadrature Optimality)**.** *An n-point quadrature rule has maximum possible degree of precision $2n-1$ if and only if the nodes are the zeros of the n-th orthogonal polynomial with respect to the weight function.*

4.2 Gauss-Legendre Quadrature

Definition. Gauss-Legendre Quadrature

For the interval $[-1,1]$ with weight function $w(x)=1$, the nodes are zeros of the Legendre polynomial $P_n(x)$, and:

$$\int_{-1}^{1} f(x)dx \approx \sum_{i=1}^{n} w_i f(x_i)$$

where $w_i = \frac{2}{(1-x_i^2)[P_n'(x_i)]^2}$.

Theorem 4.2 (Gauss-Legendre Error)**.** *If $f \in C^{2n}[-1,1]$, then:*

$$\int_{-1}^{1} f(x)dx - \sum_{i=1}^{n} w_i f(x_i) = \frac{2^{2n+1}(n!)^4}{(2n+1)[(2n)!]^3} f^{(2n)}(\xi)$$

for some $\xi \in (-1,1)$.

4.3 Other Gaussian Quadrature Rules

- **Gauss-Chebyshev**: Weight $w(x) = 1/\sqrt{1-x^2}$ on $[-1, 1]$
- **Gauss-Laguerre**: Weight $w(x) = e^{-x}$ on $[0, \infty)$
- **Gauss-Hermite**: Weight $w(x) = e^{-x^2}$ on $(-\infty, \infty)$

Intuition: Why Gaussian Quadrature Works

By choosing nodes optimally (as zeros of orthogonal polynomials), Gaussian quadrature maximizes the degree of precision. This means it integrates polynomials exactly up to degree $2n - 1$, making it extremely efficient for smooth functions that can be well-approximated by polynomials.

5 Adaptive Quadrature

Adaptive methods automatically adjust the step size to achieve desired accuracy.

5.1 Error Estimation

Definition. Richardson Extrapolation for Integration

If $Q(h)$ is a quadrature approximation with step size h and error $O(h^p)$, then:

$$Q_{ext} = \frac{2^p Q(h/2) - Q(h)}{2^p - 1}$$

provides a higher-order approximation.

5.2 Adaptive Simpson's Rule

Definition. Adaptive Simpson's Algorithm

1. Compute S_1 = Simpson's rule on $[a, b]$
2. Compute S_2 = Simpson's rule on $[a, c]$ and $[c, b]$ where $c = (a + b)/2$
3. If $|S_1 - S_2| < 15\epsilon$, accept $S_2 + (S_2 - S_1)/15$
4. Otherwise, recursively apply to $[a, c]$ and $[c, b]$ with tolerance $\epsilon/2$

6 Romberg Integration

Romberg integration systematically applies Richardson extrapolation to the trapezoidal rule.

6.1 Romberg Algorithm

Definition. Romberg Table

Define $R_{k,0} = T_{2^k}$ (composite trapezoidal with 2^k intervals) and:

$$R_{k,j} = \frac{4^j R_{k,j-1} - R_{k-1,j-1}}{4^j - 1}$$

Theorem 6.1 (Romberg Convergence)**.** *For sufficiently smooth functions, $R_{k,j}$ has error $O(h^{2j+2})$ where $h = (b-a)/2^k$.*

7 Multidimensional Integration

Extending integration to multiple dimensions presents significant challenges.

7.1 Product Rules

Definition. Tensor Product Quadrature

For integrals over rectangles:

$$\int_a^b \int_c^d f(x,y)dydx \approx \sum_{i=1}^m \sum_{j=1}^n w_i v_j f(x_i, y_j)$$

where (w_i, x_i) and (v_j, y_j) are one-dimensional quadrature rules.

The number of function evaluations grows exponentially with dimension (curse of dimensionality).

7.2 Monte Carlo Integration

Definition. Monte Carlo Integration

Approximate the integral using random sampling:

$$\int_D f(x)dx \approx \frac{|D|}{N} \sum_{i=1}^N f(x_i)$$

where x_i are random points in domain D and $|D|$ is the volume of D.

Theorem 7.1 (Monte Carlo Error)**.** *The Monte Carlo error is $O(N^{-1/2})$ regardless of dimension, making it attractive for high-dimensional problems.*

8 Special Integration Problems

8.1 Singular Integrands

For integrands with singularities, special techniques are needed:

- **Singularity subtraction**: Remove the singular part analytically
- **Variable transformation**: Transform to remove the singularity
- **Specialized quadrature**: Use rules designed for specific singularities

8.2 Oscillatory Integrands

For highly oscillatory functions:

Definition. Filon's Method

For integrals of the form $\int_a^b f(x)e^{i\omega x}dx$ with large ω, use:

$$\int_a^b f(x)e^{i\omega x}dx \approx \sum_j w_j f(x_j)e^{i\omega x_j}$$

where weights w_j are chosen to account for the oscillatory behavior.

Theorem 8.1 (Stationary Phase Method)**.** *For integrals $\int_a^b f(x)e^{i\omega g(x)}dx$ with large ω, the main contribution comes from stationary points where $g'(x) = 0$.*

Practice Problems

1. Derive the error formula for the midpoint rule using Taylor expansion.
2. Prove that Simpson's rule has degree of precision 3 (integrates cubics exactly).
3. Implement adaptive Simpson's rule and test it on functions with varying smoothness.
4. Show that the n-point Gauss-Legendre rule integrates polynomials of degree $2n - 1$ exactly.
5. Compare the efficiency of Gaussian quadrature vs. Newton-Cotes rules for integrating e^x on $[0, 1]$.

Solution:

Problem 1: Using Taylor expansion around the midpoint $c = (a + b)/2$: $f(x) = f(c) + f'(c)(x - c) + \frac{f''(\xi)}{2}(x - c)^2$. Integrating gives the midpoint rule plus error term $\frac{(b-a)^3}{24}f''(\xi)$.

Problem 2: Simpson's rule integrates $1, x, x^2, x^3$ exactly by construction (interpolating polynomial of degree 2), but fails for x^4 due to the specific coefficient structure.

Problem 3: Adaptive Simpson's rule automatically refines regions where the function changes rapidly, achieving specified tolerance with minimal function evaluations.

Problem 4: The Gauss-Legendre nodes are chosen as zeros of $P_n(x)$, ensuring that the quadrature rule integrates all polynomials up to degree $2n - 1$ exactly through the orthogonality properties of Legendre polynomials.

Problem 5: For smooth functions like e^x, Gaussian quadrature typically requires far fewer function evaluations than Newton-Cotes rules to achieve the same accuracy, due to its optimal node placement.